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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:55:27",
  "variant" : 2,
  "subjectName" : "operating-systems",
  "categories" : [ "Memory Management", "Kernel Architecture", "Deadlocks & Synchronization" ],
  "questions" : [ {
    "question" : "What is a page fault?",
    "answers" : [ "A physical hardware failure inside RAM modules", "A data anomaly where two files
share the same sector", "A crash caused by syntax compile errors", "An interrupt triggered when a
program accesses a page not currently mapped in RAM" ],
    "correctIndex" : 3
  }, {
    "question" : "What is the role of a Translation Lookaside Buffer (TLB)?",
    "answers" : [ "To act as a specialized cache for virtual-to-physical address translations", "To
handle network data package buffering streams", "To map file allocation records sequentially on
disk", "To store execution metrics logs for background tasks" ],
    "correctIndex" : 0
  }, {
    "question" : "What is the function of the Memory Management Unit (MMU)?",
    "answers" : [ "To allocate network sockets to background daemons", "To partition hard drive
sectors dynamically at runtime", "To map virtual addresses generated by the CPU to physical
memory locations", "To clear cache registers when a context switch occurs" ],
    "correctIndex" : 2
  }, {
    "question" : "What privilege layer does the operating system kernel run in on modern CPUs?",
    "answers" : [ "Hypervisor mode", "Safe mode", "Kernel mode (Ring 0)", "User mode (Ring 3)" ],
    "correctIndex" : 2
  }, {
    "question" : "What is the primary purpose of a system call?",
    "answers" : [ "To compile source code files into machine code", "To provide an interface for user
programs to request services from the kernel", "To clear CPU cache configurations automatically",
"To manage graphical window rendering pipelines" ],
    "correctIndex" : 1
  }, {
    "question" : "What characterizes a monolithic kernel architecture?",
    "answers" : [ "It runs exclusively on virtual machines", "All operating system services run inside a
single privileged kernel space", "OS services are moved to separate user-space modules", "It
eliminates context switching overhead entirely" ],
    "correctIndex" : 1
  }, {
    "question" : "What is a binary semaphore commonly called when used strictly as a locking

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mechanism?",
"answers" : [ "Critical Section", "Mutex", "Condition Variable", "Monitor" ],
"correctIndex" : 1
}, {
"question" : "What classical algorithm can be utilized to safely avoid deadlocks in resource
allocation?",
"answers" : [ "Round Robin Algorithm", "Banker's Algorithm", "De Morgan's Optimization",
"Dijkstra's Shortest Path" ],
"correctIndex" : 1
}, {
"question" : "What condition occurs when multiple threads modify shared data concurrently with
output depending on execution order?",
"answers" : [ "Livelock loop", "Race Condition", "Deadlock state", "Starvation anomaly" ],
"correctIndex" : 1
}, {
"question" : "Which of the following is NOT one of the Coffman conditions required for a
deadlock?",
"answers" : [ "Mutual Exclusion", "Circular Wait", "Preemption", "Hold and Wait" ],
"correctIndex" : 2
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}
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